

Improving Admittance Control on Wuji Hand

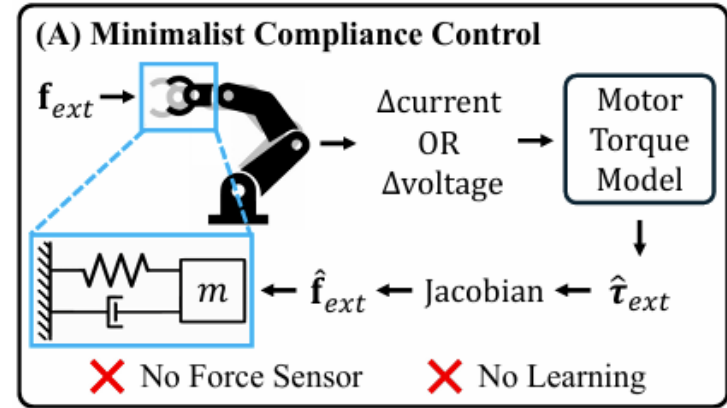
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Introduction

- Admittance control converts external force into finger displacement by modeling the system as a virtual mass–spring–damper.

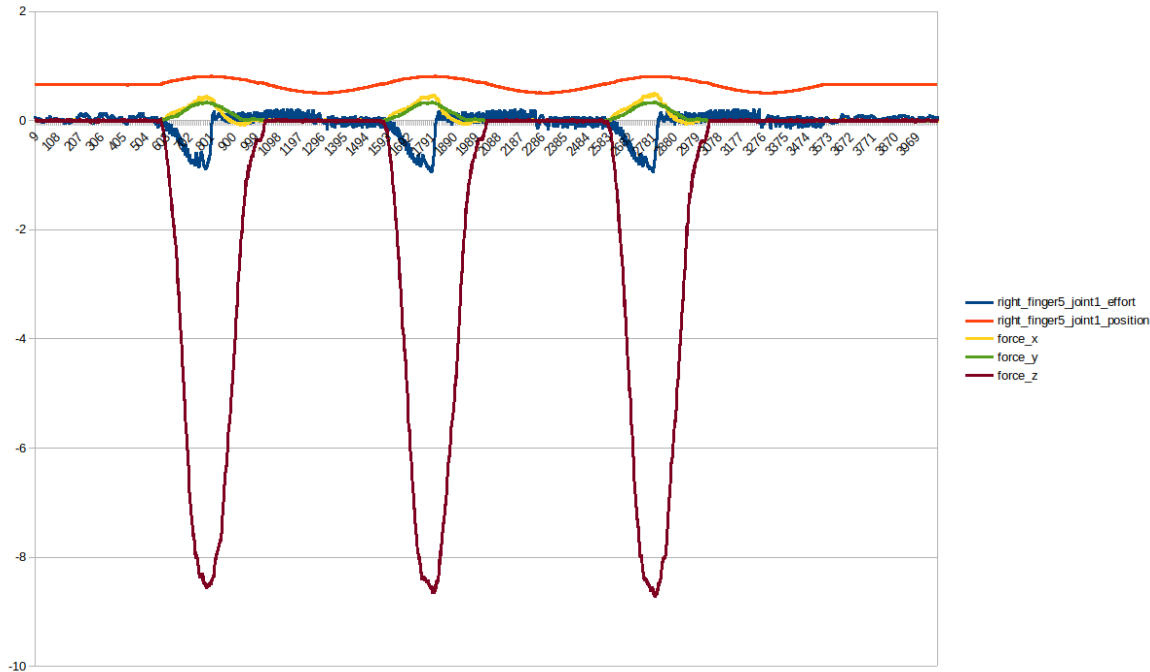
- $M\Delta\ddot{x} + D\Delta\dot{x} + K\Delta x = F_{\text{ext}}$
- $x_{\text{new}} = x_{\text{nominal}} + \Delta x$



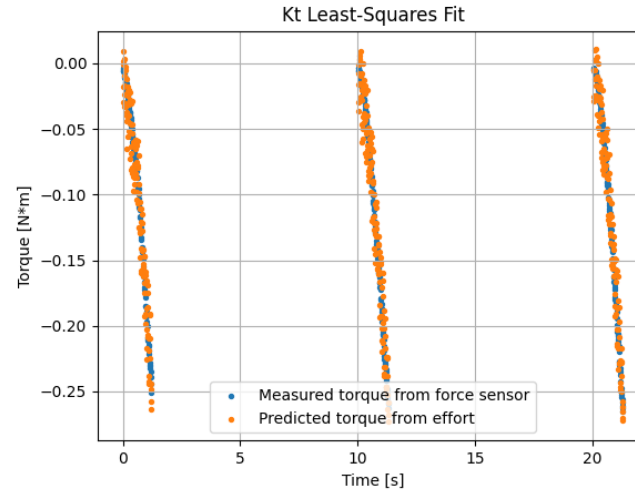
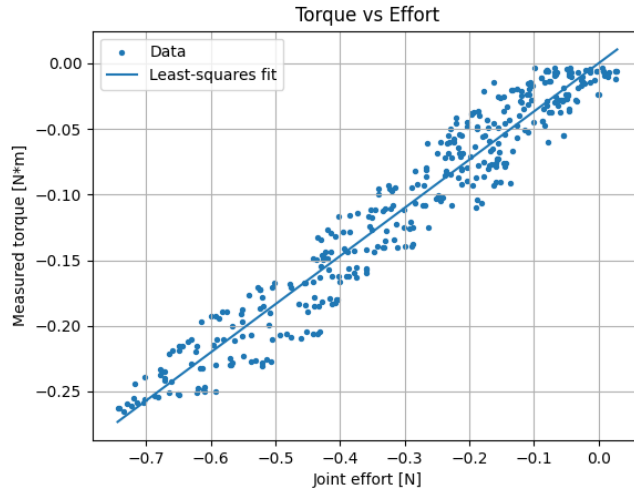
- Currently using current-based force estimation as input to the admittance control
- $\tau = K_t I_w$

System Identification

- 3D printed test fixtures
- Generated sine wave motion on Wuji hand for each joint



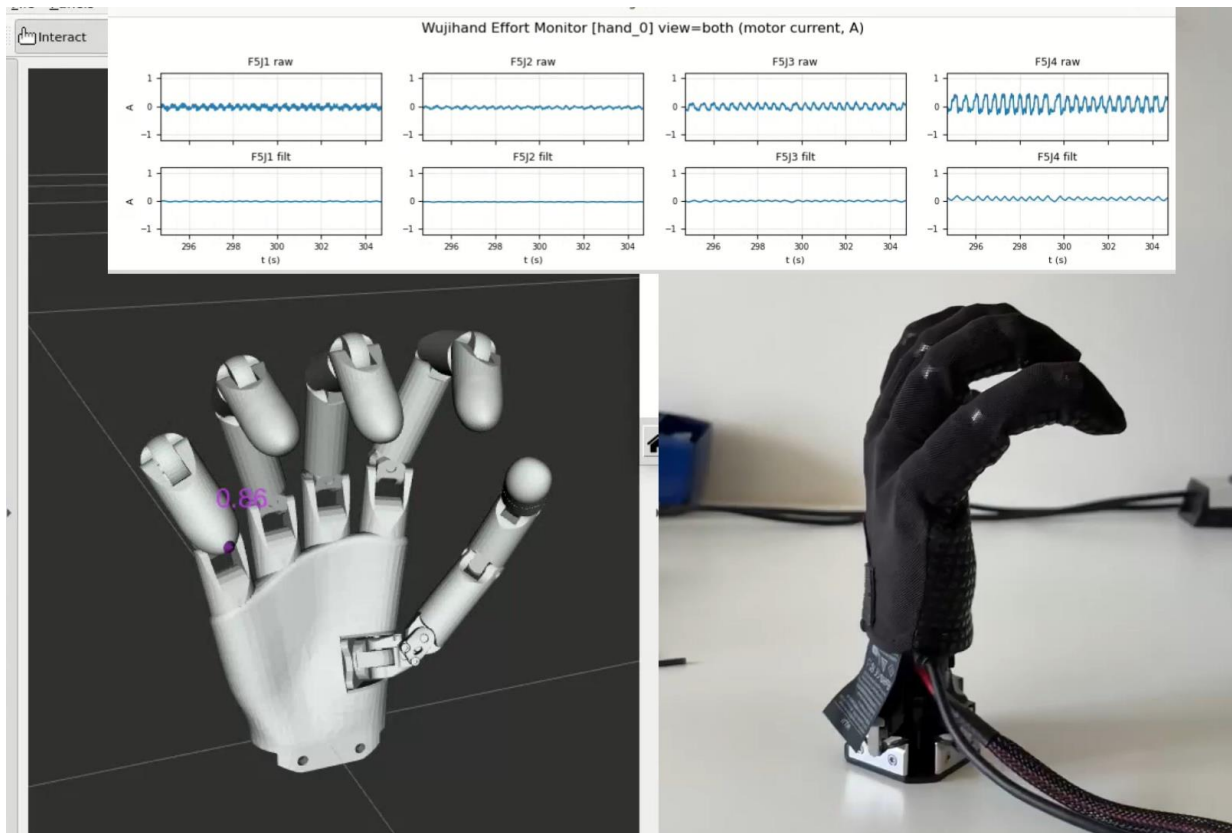
Results



Kt Analysis:

Joint	Kt	RMSE [N·m]	NRMSE [%]
F5 J1/J3/J4	0.3647	0.0190	8.11
F5 J2	0.1991	0.0118	9.82

Improved Force Estimation and Admittance Control



Next Steps / Future Direction

- Identify root cause of oscillatory behaviour during admittance control
- Tune admittance parameters for smoother, compliant behaviour
- Integrate tactile glove force input
- Build transformation function algorithm to transform glove finger force to palm frame